

## **Exp No: 12 EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS**

### **AIM:**

To write an Embedded C program to multiply two 16-bit numbers using Keil software

### **SOFTWARE REQUIRED:**

- Keil

### **Program:**

```
#include <reg51.h>

void main(void)
{
    unsigned int num1, num2;
    unsigned long int product;

    while(1)
    {
        num1 = 0x2222;
        num2 = 0BBBB;

        product = (unsigned long int)num1 * (unsigned long int)num2;

        P0 = (unsigned char)(product & 0xFF);
        P1 = (unsigned char)((product >> 8) & 0xFF);
        P2 = (unsigned char)((product >> 16) & 0xFF);
        P3 = (unsigned char)((product >> 24) & 0xFF);
    }
}
```

## OUTPUT:

The screenshot displays the Keil uVision IDE interface during the execution of a program. The main window shows the source code for 'EXP\_12.c', which includes a 'while' loop. The 'Registers' window on the left lists various registers and their values. The 'Disassembly' window at the top shows the assembly code corresponding to the source code. The 'Call Stack + Locals' window on the right shows the current stack frame with variables 'num1' (0x2222), 'num2' (0xB8BB), and 'product' (0x1907C4D6). The 'Command' window at the bottom shows the command 'Load "C:\\Keil\_v5\\CS1\\Examples\\HELLO\\Objects\\EXP\_12"'. The status bar at the bottom indicates 'Simulation' mode with a timer of 944.25102850 sec and a clock of 11:43 PM on 8/7/2026.

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## RESULT:

Thus the program has been successfully verified and executed.